

a

170 bp sequence

count occurrence of all possible 6-mers

linear model

predict $\log_2(\text{enhancer strength})$

AGTGAGCTGT ... ACCTGCAGGG →

kmer-1	kmer-2	...	kmer-4095	kmer-4096
AAAAAA	AAAAAC	...	TTTTTG	TTTTTG
1	0	...	0	3

→

linear regression

$$\log_2(\text{enhancer strength}) = \beta_0 + \beta_1 \text{kmer-1} + \beta_2 \text{kmer-2} + \dots + \beta_{4095} \text{kmer-4095} + \beta_{4096} \text{kmer-4096}$$

→

light	dark	warm	cold	maize
0.24	0.32	0.24	0.26	3.57

b